TM Forum Specification

Product Usage Management User Guide

TMF635

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Introduction

This document is the user guide of the REST API for Product Usage Management. It includes the data model definition as well as all available operations for the ProductUsage resource.

The Product Usage Management API provides a standardized mechanism for Product Usage management such as the creation, update, retrieval of a Product Usage or of a collection of Product Usage.

A Product Usage entity represents the structured report of a service that has been delivered, guided to a product, used and is considered billable. Each Product Usage entity is defined by a corresponding Product Usage Specification, which standardizes what type of usage occurred (e.g., voice, data session, API call) and its associated characteristics (e.g., duration, data consumption, correlation identifier). The main content of a Product Usage is a list of characteristics (name-value pairs), that are needed to guide and/or to rate the usage. For example, A and B numbers in a telephone call, amount of data transferred in a data packet transfer, etc. Other key attributes of Product Usage are its id, creation date, status and the id of the related product and of the related Product Usage Specification.

TMF767 Product Usage Catalog Management API is responsible for managing Product Usage Specifications that represents a catalog view of the targeted Product Usage comprised of characteristics specifications which define all the attributes known for the particular usage.

The Product Usage API manages both rated and non-rated usages. For example, it allows a service provider:

- To retrieve usage generated by a partner platform in order to rate it
- To provide rated usages to a partner for their consumption

A ProductUsage represents an entity that may be of interest to the business and can have charges applied to it. It is comprised of characteristics which define all the attributes known for that particular type of usage. The main attributes are its id, creationDate, usageType, related UsageSpecification, characteristics (name/value pairs), status, price charged (date, amount, taxes), product involved in the usage. A ProductUsage follows a simple life cycle which is presented in the following chapter 'ProductUsage life cycle'.

The Product Usage API exposes the following operations on the ProductUsage resource:

- Creation of a ProductUsage. Note that creation of a ProductUsage can be done without the API, rather as the result of internal processing by an online charging system or policy control system
- Retrieval of a ProductUsage or a collection of ProductUsage, depending on filter criteria
- Partial update of a ProductUsage or of a collection of ProductUsage
- Deletion of a ProductUsage

The Product Usage API also optionally exposes the ExportJob resource:

- An ExportJob represents a task used to export productUsage representations.

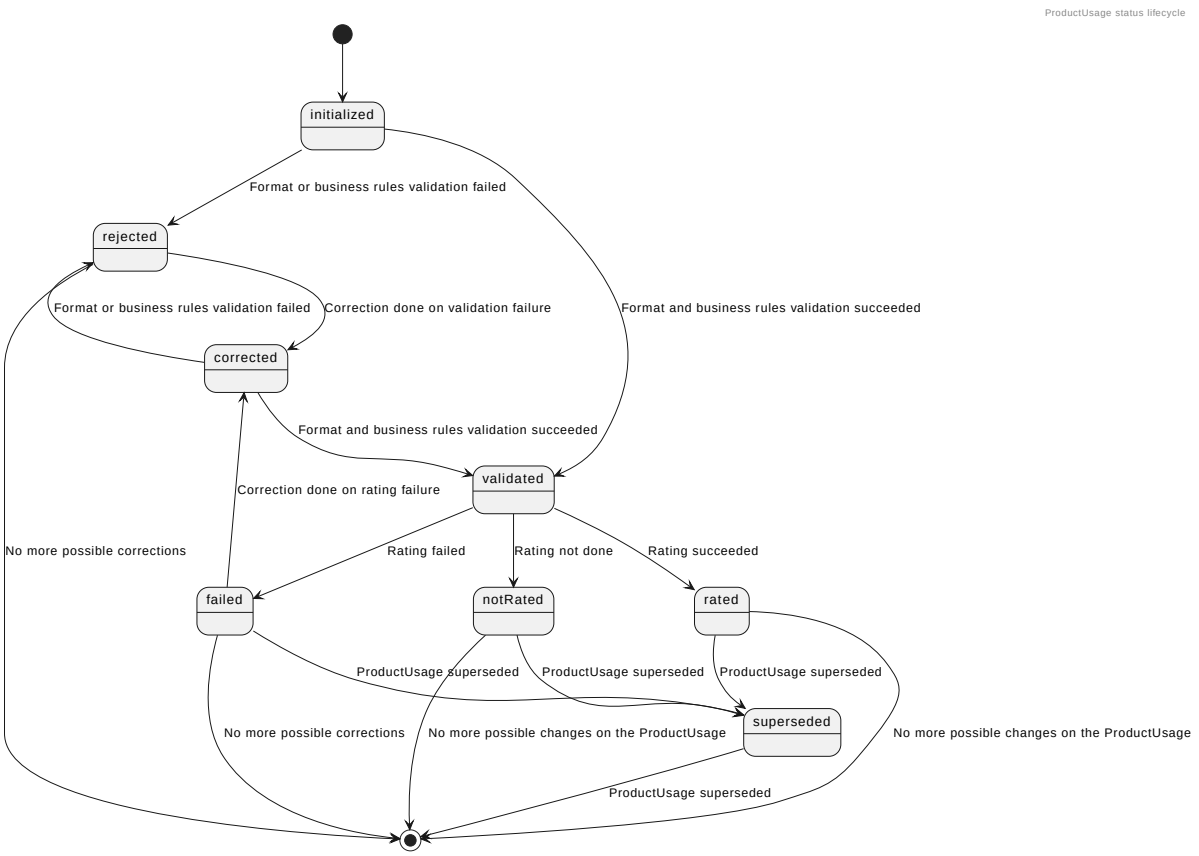
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- To trigger the export of productUsage, an ExportJob is created by the API client using POST operation on the ExportJob resource.
- The ExportJob is then processed by the API server, which generates a file containing the productUsage representations and stores it in a designated location.
- The API client can then retrieve the ExportJob status and the generated file using the GET operation on the ExportJob resource.
- The ExportJob can be used to export productUsage representations in various formats, such as JSON or CSV.
- The ExportJob resource allows the API client to specify the filter criteria for the productUsage representations to be exported, such as creationDate range, status, related product.id.
- An ExportJob follows a simple life cycle which is presented in the following chapter 'ExportJob life cycle'.

The Product Usage API exposes the following operations on the ExportJob resource:

- Creation of an ExportJob
- Retrieval of an ExportJob or a collection of ExportJob, depending on filter criteria
- Deletion of an ExportJob

ProductUsage life cycle



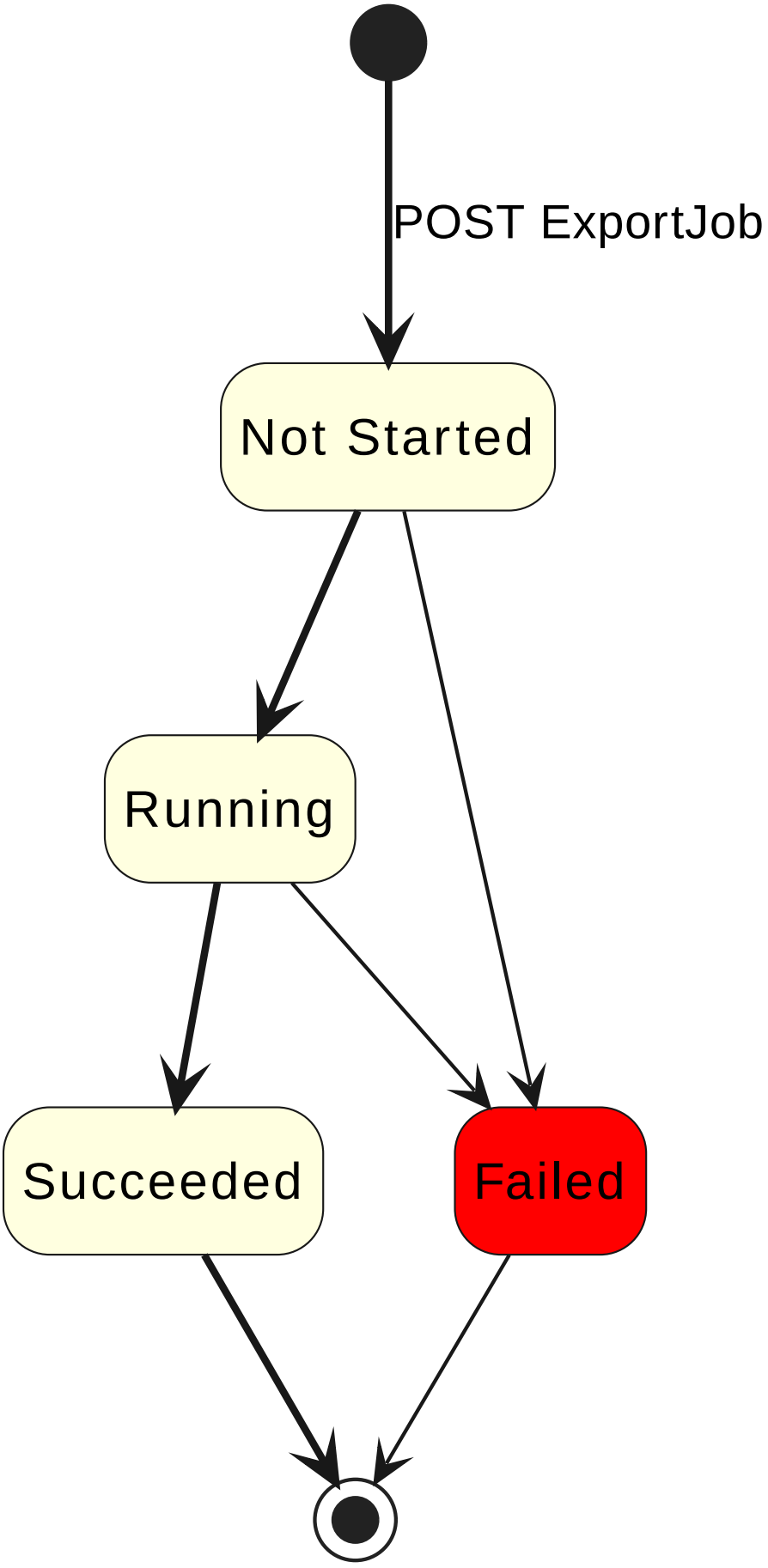
State	Definition

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initialized	This acknowledges that the ProductUsage has been initialized and is pending validation
rejected	The rejected state is where invalid information is provided through the ProductUsage instance (not necessarily checked synchronously)
validated	The product, customer billing account and pricing rules used to rate the ProductUsage have been found
rated	The ProductUsage has been rated, at least one product price has been created (usageProductPrice) within the ProductUsage
corrected	The ProductUsage is resubmitted after error correction
superseded	This ProductUsage record has been superseded by another due to error (e.g., wrong price) and is no longer billable
notRated	The ProductUsage has not been rated, due to business rules (e.g., no price applicable for this usage type)
failed	The ProductUsage record has been validated but rating failed

ExportJob life cycle

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The ExportJob task can be either:

- short-running task, where the result of the task can be returned in the same call to the client
- long-running task, where the result cannot be computed immediately; then the task resource can be created and returned to the client in the same API call

When an implementation for a Task supports both alternatives, it MAY decide at runtime to execute the task as short-running or long-running task, taking into consideration its expectation of the duration of task execution, e.g. by investigating any request parameters or payload values.

In that case, the consumer must be able to distinguish and handle both variants of the task operation.

Refer to TMF630 REST guidelines for more details on Task pattern.

Information Framework

This Product Usage API manages an entity that is described in Information Framework (SID) in Product Usage ABE as ProductUsage entity.

These links permit to retrieve Information Framework documentation regarding Product Usage domain :

- Usage Specification : https://www.tmforum.org/MODA/EARoot/EA4/EA2/EA4/EA6/EA1712.htm
- Product Usage Part 1 : https://www.tmforum.org/MODA/EARoot/EA4/EA2/EA4/EA6/EA1716.htm
- Product Usage Part 2 : https://www.tmforum.org/MODA/EARoot/EA4/EA2/EA4/EA6/EA1718.htm

Information Framework excerpts:

Put simply, usage is how much service is used, by whom is it used, where and when is it used and circumstances under which it is used.

The ProductUsage describes the usage of an instantiated Product.

Note that the API representation of Information Framework model is partial. In particular the direct relationship between a ProductUsage (that would be originated from a ResourceUsage) and a ResourceUsage is not implemented for now because of lack of business need for this.

Note that Information Framework and API specification may evolve independently but tend to be consistent as much as possible.

ODA Components

This Product Usage API is exposed by TMFC040 Product Usage Management.

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Note that ODA Component and API specification may evolve independently but tend to be consistent as much as possible.

Future evolutions

Future evolutions of this API may take into account possible evolutions of Information Framework Product Usage ABE and of TMFC040 Product Usage Management, in particular the rated price that could be externalised in a separate entity.

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Sample Use Cases

The examples in this User Guide and OAS file come from the TMFS005 Use Case Billing Fiber Contract and TMFS009 Use Case Usage and Balance Management documents.

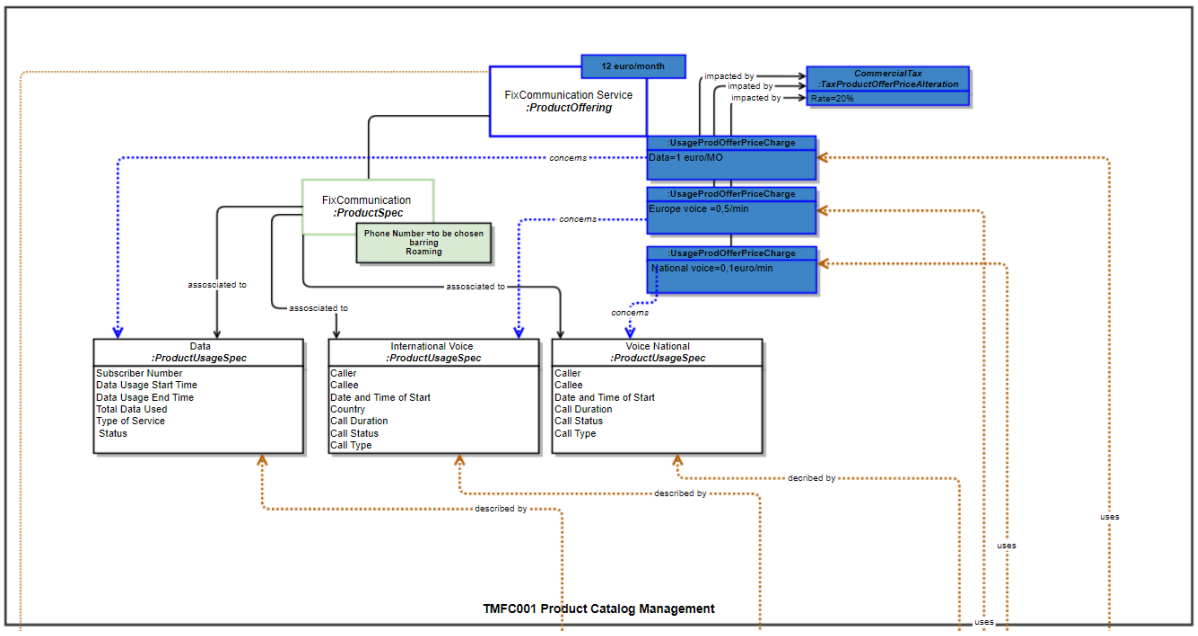
- Creation of a simple ProductUsage relating to a completed and rated voice call retrieval
- listing ProductUsage of usageType “InternationalVoice”
- retrieving a specific ProductUsage based on its id
- updating a Product Usage status to "recycled"

Note that entities and characteristic names don’t always fully match but the meaning should match.

Catalog view

This excerpt from TMFS005 presents the catalog view of a Fix Communication offering.

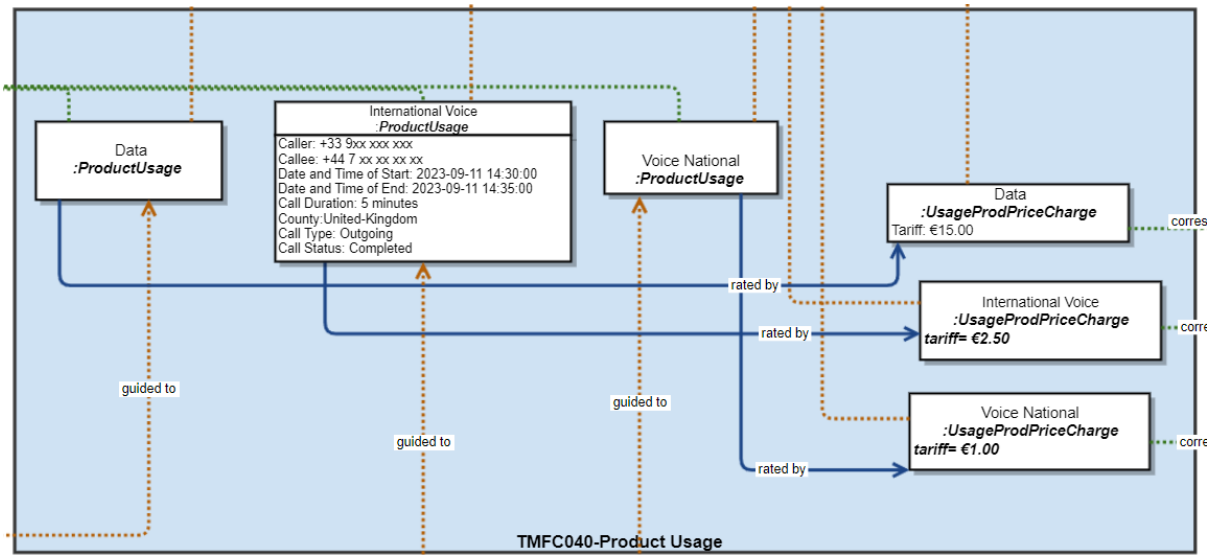
The Fix Communication ProductSpecification has 3 ProductUsageSpecification : National Voice, International Voice and Data. These ProductUsageSpecification are under the responsibility of the Product Catalog component. They can be created and updated by a employee of a marketing team having access to the Product Catalog administration software.



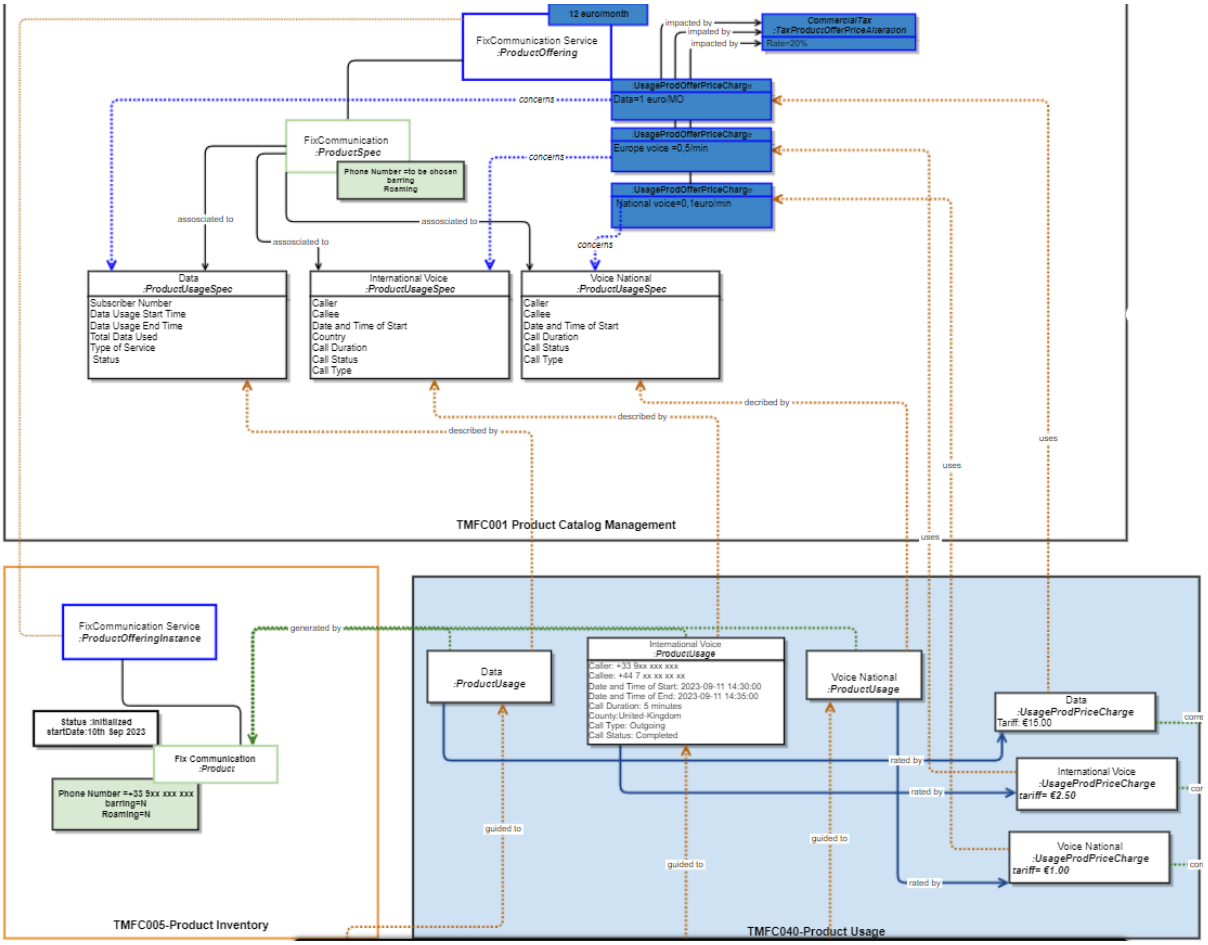
Inventory view

This excerpt from TMFS005 presents the ProductUsage representation once created and rated.

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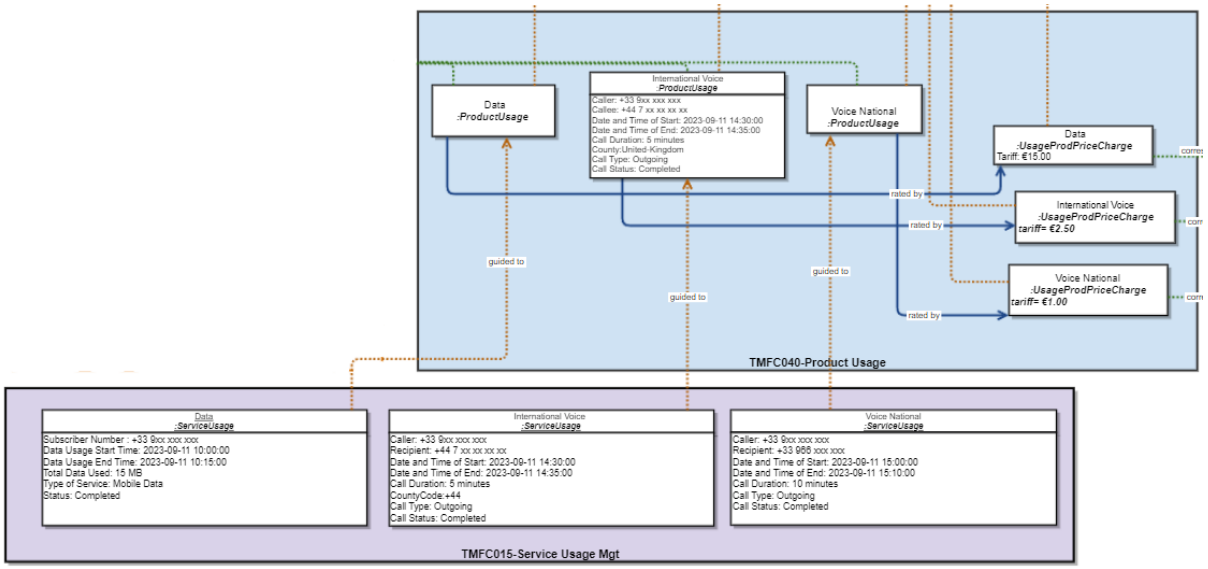


This view permits to see the relationship between the inventory view of the Fix Communication product and its associated ProductUsage created after a voice call has been completed, and also the relationship between the ProductUsage and their corresponding ProductUsageSpecification, same for UsageProductPrice and ProductOfferingPrice.



These ProductUsage are created based on preliminary ServiceUsage entities that contain characteristics of the voice call.

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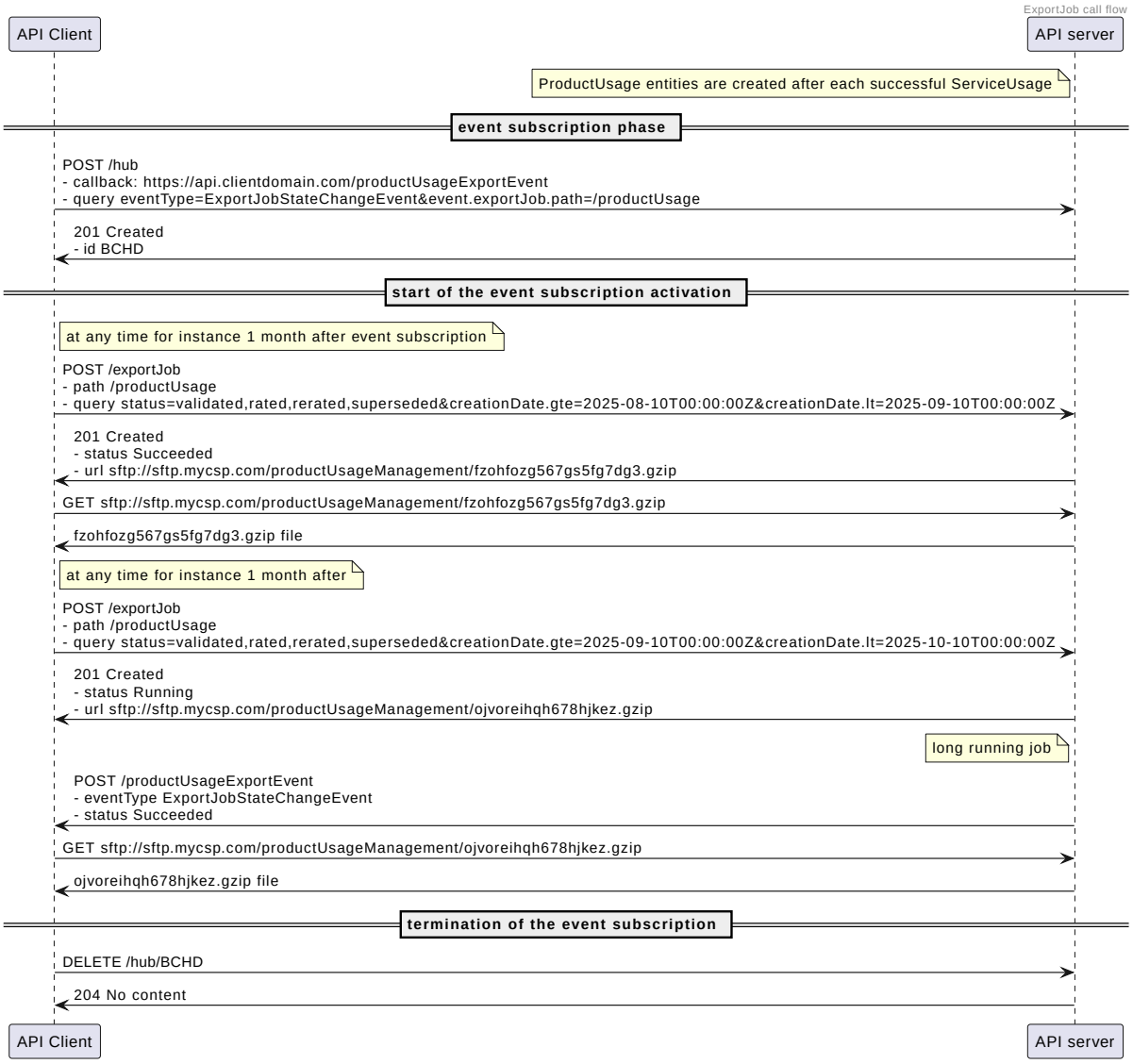


Export

Using ExportJob, the ProductUsage can be exported to a file.

Note : an implementation of this API could expose only exportJob resource if it's agreed between API client and API server that the exportJob is the only way to retrieve ProductUsage (indivivual GET queries of productUsage not allowed). This can be useful for performance reasons, e.g. if the producUsage are of a high number and/or if the interface explicitely allows only 1 single mechanism to retrieve the productUsage entities for simplicity reasons.

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a StringCharacteristic and BooleanCharacteristic inheriting properties from the abstract Characteristic entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Characteristic instances some may be instances of StringCharacteristic where other could be instances of BooleanCharacteristic. The @type gives this information. All resources and sub-resources of the Product Usage API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance a ProductUsageSpecificationtRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as ProductUsageSpecificationtRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

Managed Entity and Task Resource Models

ProductUsage resource

An occurrence of usage on a product, typically derived from one or various service usages.

Resource model

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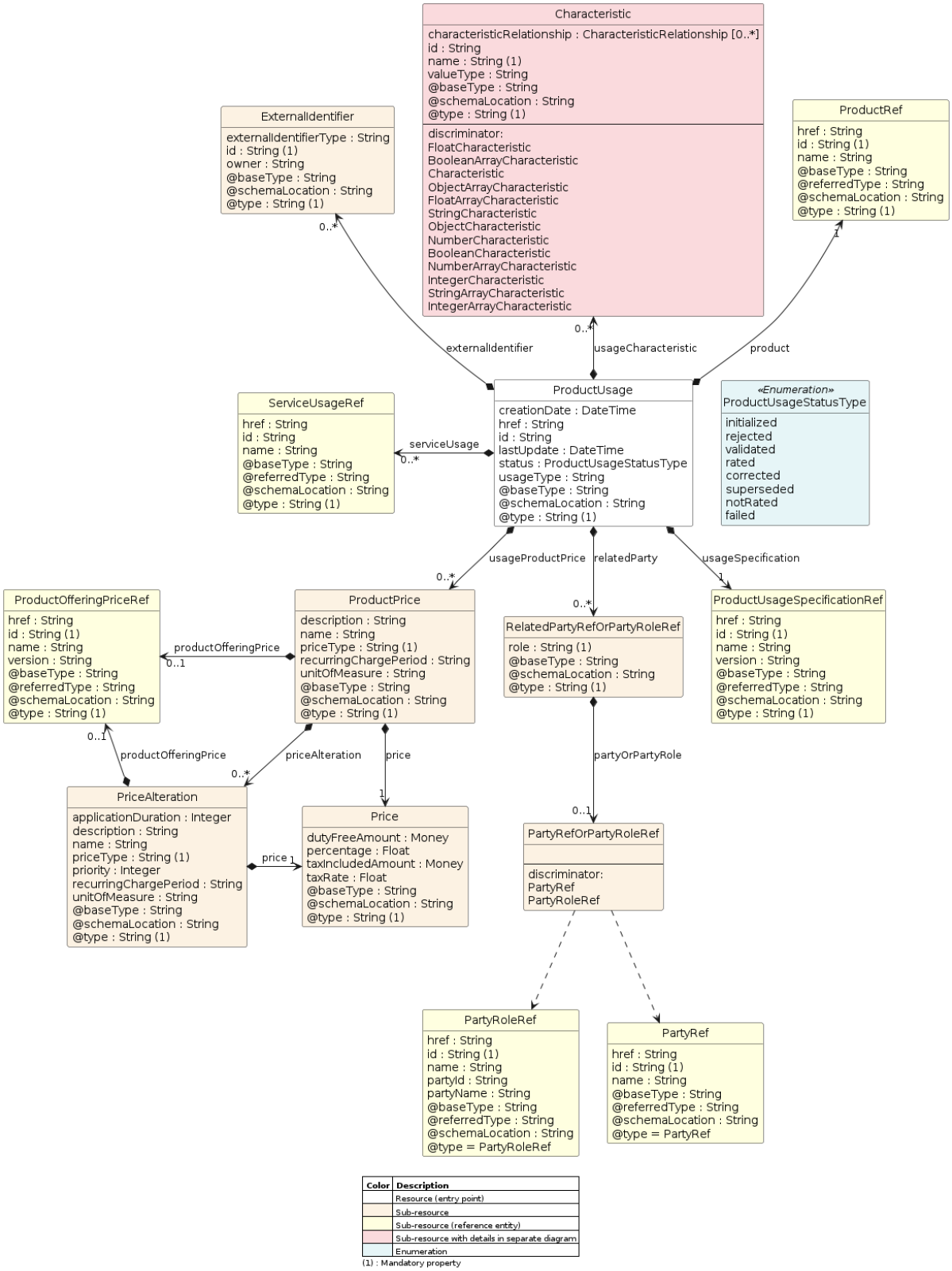


Figure 1 - ProductUsage

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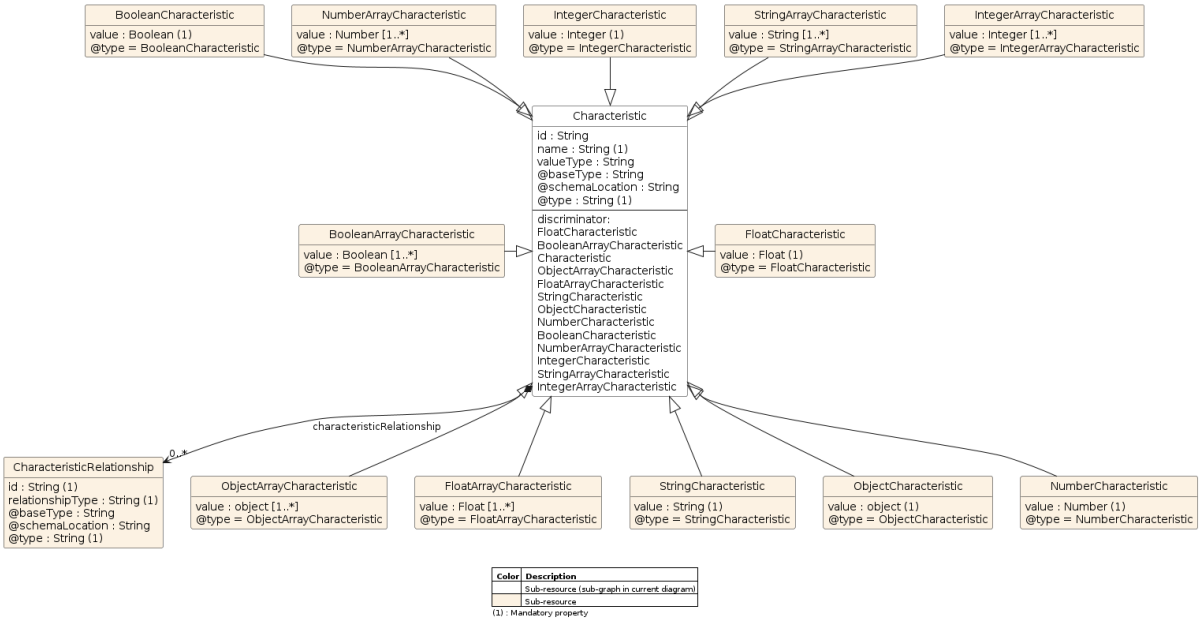


Figure 2 - Characteristic

Field descriptions

ProductUsage fields

creationDate	A DateTime. Creation date of the product usage.
externalIdentifier	An ExternalIdentifier. A list of external identifiers associated with this product usage.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
lastUpdate	A DateTime. The date and time that the product usage was last updated.
product	A ProductRef. A Product reference.
relatedParty	A RelatedPartyRefOrPartyRoleRef. Parties related to the product usage.
serviceUsage	A ServiceUsageRef. The service usage(s) that participated in the creation of this product usage.
status	A ProductUsageStatusType. Possible values for the status of the Product Usage. ENUMERATED with values: * initialized * rejected * validated * rated * corrected * superseded * notRated * failed

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usageCharacteristic	A Characteristic. Characteristics of product usage. Each characteristic should correspond to a characteristic specification described in the corresponding product usage specification. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
usageProductPrice	A ProductPrice. Price assigned to the product usage when product usage is rated. The price should be valued without applied taxes (dutyFreeAmount), that permits to avoid loss of precision. Better precision is reached by applying taxes on the total dutyFreeAmount of several Product Usage Price. Note that in this API the recurringChargePeriod attribute is not applicable.
usageSpecification	A ProductUsageSpecificationRef. ProductUsageSpecification reference. ProductUsageSpecification is a detailed description of a product usage event that are of interest to the business and can have charges applied to it. It is comprised of characteristics, which define all attributes known for a particular type of usage.
usageType	A String. This attribute can be used to precise the type of usage, for instance in the case 1 single ProductUsageSpecification manages different types of usages.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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ProductRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.

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partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceUsageRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. The name of the usage.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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	Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

CharacteristicRelationship sub-resource fields

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. A characteristic which value is an array of Boolean(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.

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name	A String. Name of the characteristic.
value	A Boolean. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. A characteristic which value is an array of Float(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.

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id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. A characteristic which value is an array of Integer(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. A characteristic which value is an array of Number(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

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characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Collection of characteristic values.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

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characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Collection of string characteristics.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductPrice sub-resource fields

description	A String. Description of the Product price.
name	A String. Name of the Product price.
price	A Price. Provides all amounts (tax included, duty free, tax rate), used currency and percentage to apply for Price and Price Alteration.
priceAlteration	A PriceAlteration. Collection of price changes.
priceType	A String. Indicate if the price is for recurrent or no-recurrent charge.
productOfferingPrice	A ProductOfferingPriceRef. ProductPriceOffering reference. An amount, usually of money, that is asked for or allowed when a ProductOffering is bought, rented, or leased.
recurringChargePeriod	A String. Used for recurring charge to indicate period (month, week, etc..).

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unitOfMeasure	A String. Unit of Measure if price depending on it (Gb, SMS volume, etc..).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Price sub-resource fields

dutyFreeAmount	A Money. A base / value business entity used to represent money.
percentage	A Float. Percentage to apply for ProdOfferPriceAlteration.
taxIncludedAmount	A Money. A base / value business entity used to represent money.
taxRate	A Float. Tax rate.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PriceAlteration sub-resource fields

applicationDuration	An Integer. Duration during which the alteration applies on the order item price (for instance 2 months free of charge for the recurring charge).
description	A String. A narrative that explains in detail the semantics of this order item price alteration.
name	A String. Name of the order item price alteration.
price	A Price. Provides all amounts (tax included, duty free, tax rate), used currency and percentage to apply for Price and Price Alteration.
priceType	A String. A category that describes the price such as recurring, one time and usage.
priority	An Integer. Priority level for applying this alteration among all the defined alterations on the order item price.
productOfferingPrice	A ProductOfferingPriceRef. ProductPriceOffering reference. An amount, usually of money, that is asked for or allowed when a ProductOffering is bought, rented, or leased.
recurringChargePeriod	A String. Could be month, week…
unitOfMeasure	A String. Could be minutes, GB…
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOfferingPriceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Version of the product offering price.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOfferingPriceRef sub-resource fields

See [ProductOfferingPriceRef] for details.

ProductUsageSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Version of the product usage specification.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'ProductUsage' resource object.

```
{
  "@type": "ProductUsage",
  "href": "https://mycsp.com/tmf-
```

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```
api/productUsageManagement/v5/productUsage/83c780b5-517c-4e28-9924-625e3fc7b341",
  "id": "83c780b5-517c-4e28-9924-625e3fc7b341",
  "creationDate": "2023-09-11T14:35:05Z",
  "lastUpdate": "2023-09-11T14:38:00Z",
  "status": "rated",
  "usageType": "InternationalVoice",
  "usageCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "name": "originatingCountryCode",
      "value": "FR"
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationCountryCode",
      "value": "GB"
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationPhoneNumber",
      "value": "+447092734896"
    },
    {
      "@type": "StringCharacteristic",
      "name": "originatingPhoneNumber",
      "value": "+33939487561"
    },
    {
      "@type": "StringCharacteristic",
      "name": "callType",
      "value": "outgoing"
    },
    {
      "@type": "IntegerCharacteristic",
      "name": "duration",
      "value": 300
    },
    {
      "@type": "StringCharacteristic",
      "name": "startDateTime",
      "valueType": "date-time",
      "value": "2023-09-11T14:30:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "endDateTime",
      "valueType": "date-time",
      "value": "2023-09-11T14:35:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "callStatus",
      "value": "completed"
    }
  ],
  "usageProductPrice": [
    {
      "@type": "ProductPrice",
      "price": {
        "@type": "Price",
        "dutyFreeAmount": {
          "unit": "EUR",
          "value": 2.5
```


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```
        },
        "taxRate": 20
      },
      "productOfferingPrice": {
        "@type": "ProductOfferingPriceRef",
        "@referredType": "ProductOfferingPrice",
        "href": "https://mycsp.com/tmf-
api/productCatalogManagement/v5/productOffering/419ce511-dbb9-4753-8155-
3391b9998cd7",
        "id": "419ce511-dbb9-4753-8155-3391b9998cd7"
      },
      "priceType": "usage"
    }
  ],
  "serviceUsage": [
    {
      "@type": "ServiceUsageRef",
      "@referredType": "ServiceUsage",
      "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsage/472999bf-1aeb-48e6-b954-1303c2c7939b",
      "id": "472999bf-1aeb-48e6-b954-1303c2c7939b"
    }
  ],
  "product": {
    "@type": "ProductRef",
    "@referredType": "Product",
    "href": "https://mycsp.com/tmf-api/productInventory/v5/product/05c89397-6984-
4c8e-9030-7cf092754f18",
    "id": "05c89397-6984-4c8e-9030-7cf092754f18",
    "name": "FixCommunication"
  },
  "usageSpecification": {
    "@type": "ProductUsageSpecificationRef",
    "@referredType": "ProductUsageSpecification",
    "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
    "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
    "name": "InternationalVoice",
    "version": "2.5"
  }
}
```


ExportJob resource

Represents a task used to export resources to a file.

Resource model

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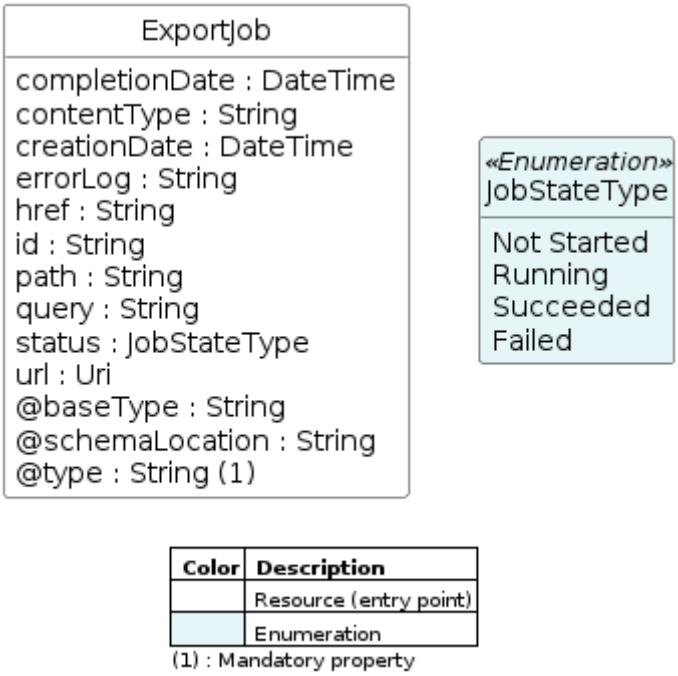


Figure 3 - ExportJob

Field descriptions

ExportJob fields

completionDate	A DateTime. Data at which the job was completed.
contentType	A String. The format of the exported data. This value should be according to the IANA standard list of media types. Default is 'application/json'.
creationDate	A DateTime. Date at which the job was created.
errorLog	A String. Path to file or stream where errors encountered during the job processing can be written.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
path	A String. A Json Pointer or a Json Path relative to the API root node that points to the resource acting as the source for streaming content to the export destination specified by the ExportJob. Defaults to the API?s root node. Using the JSON Pointer resp. JSON Path with reference to the API root guarantees that the path cannot point to a resource outside the scope of this API, e.g. to export some resource managed by a different API.
query	A String. Used to scope the exported data. Identical to GET filter construct using the API?s node defined by path as base. Defaults to requesting all resources of the node defined by the path.

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status	A JobStateType. Valid values for the state of a batch job (e.g. catalog import). ENUMERATED with values: * Not Started * Running * Succeeded * Failed
url	A Uri. URL pointing at the location for the exported file. The URL sould be of the form file://host/path where host is the fully qualified domain name of the system on which the path is accessible, and path is a hierarchical directory path of the form directory/directory/??/name. Other schemes can also be supported, e.g. 'ftp' and 'http', preferably with their TLS encrypted variants 'sftp' resp 'https'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'ExportJob' resource object.


```
{
  "@type": "ExportJob",
  "href": "https://mycsp.com/tmf-api/productUsageManagement/v5/exportJob/aa1d7102-
a2de-4e50-abc5-15cddac9c5d4",
  "id": "aa1d7102-a2de-4e50-abc5-15cddac9c5d4",
  "creationDate": "2023-10-01T00:00:20Z",
  "completionDate": "2023-10-01T00:02:30Z",
  "status": "Succeeded",
  "path": "/productUsage",
  "query": "status=validated,rated,notRated,superseded&creationDate.gte=2023-09-
10T00:00:00Z&creationDate.lt=2023-10-10T00:00:00Z",
    "contentType": "application/gzip",
    "url": "sftp://sftp.mycsp.com/productUsageManagement/fzohfozg567gs5fg7dg3.gzip"
}
```


Notification Resource Models

3 notifications are defined for this API.

Notifications related to ExportJob:

- Create Event
- Delete Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId),

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an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

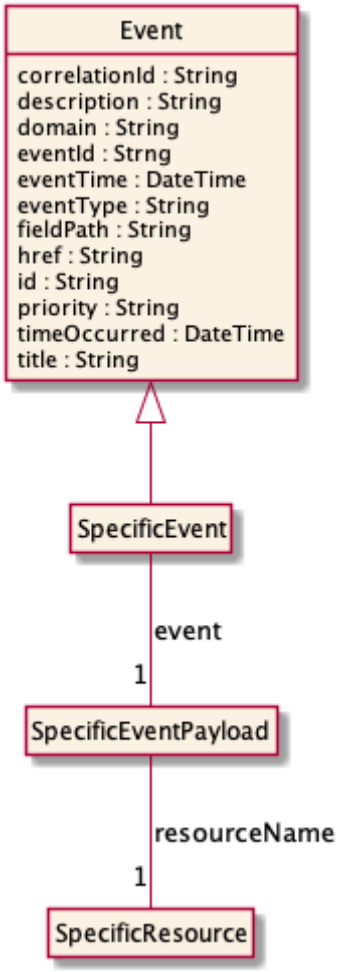


Figure 4 Notification Pattern

ExportJob

Create Event

Creating a ExportJobCreateEvent event

```
Content-Type: application/json
{
  "@type": "ExportJobCreateEvent",
  "@baseType": "Event",
  "domain": "Product Usage",
  "timeOccurred": "2023-10-01T00:00:20Z",
  "eventId": "0d4f270c-89a6-43ae-aeb9-beda67aacd70",
  "eventType": "ExportJobCreateEvent",
  "event": {
    "exportJob": {
      "@type": "ExportJob",
      "href": "https://mycsp.com/tmf-
```

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```
api/productUsageManagement/v5/exportJob/aa1d7102-a2de-4e50-abc5-15cddac9c5d4",
      "id": "aa1d7102-a2de-4e50-abc5-15cddac9c5d4",
      "creationDate": "2023-10-01T00:00:20Z",
      "status": "Running",
      "path": "/productUsage",
      "query": "status=validated,rated,notRated,superseded&creationDate.gte=2023-09-
10T00:00:00Z&creationDate.lt=2023-10-10T00:00:00Z",
      "contentType": "application/gzip",
      "url":
"sftp://sftp.mycsp.com/productUsageManagement/fzohfozg567gs5fg7dg3.gzip"
    }
  }
}
```

Delete Event

Creating a ExportJobDeleteEvent event

```
Content-Type: application/json
{
  "@type": "ExportJobDeleteEvent",
  "@baseType": "Event",
  "domain": "Product Usage",
  "timeOccurred": "2025-10-01T00:00:30Z",
  "eventId": "9ce7d74b-1ffd-4635-9ae8-bb23216717d0",
  "eventType": "ExportJobDeleteEvent",
  "event": {
    "exportJob":  {
      "@type": "ExportJob",
      "href": "https://mycsp.com/tmf-
api/productUsageManagement/v5/exportJob/aa1d7102-a2de-4e50-abc5-15cddac9c5d4",
      "id": "aa1d7102-a2de-4e50-abc5-15cddac9c5d4"
    }
  }
}
```

State Change Event

Creating a ExportJobStateChangeEvent event

```
Content-Type: application/json
{
  "@type": "ExportJobStateChangeEvent",
  "@baseType": "Event",
  "domain": "Product Usage",
  "timeOccurred": "2023-10-01T00:02:30Z",
  "eventId": "f5e40b51-6a6a-4926-848e-03a666ed0ce5",
  "eventType": "ExportJobStateChangeEvent",
  "event": {
    "exportJob": {
      "@type": "ExportJob",
      "href": "https://mycsp.com/tmf-
api/productUsageManagement/v5/exportJob/aa1d7102-a2de-4e50-abc5-15cddac9c5d4",
      "id": "aa1d7102-a2de-4e50-abc5-15cddac9c5d4",
      "status": "Succeeded"
    }
  }
```

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```
}
```

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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on ProductUsage

Retrieves a ProductUsage by ID

`GET /productUsage/{id}?fields=…`

Description

This operation retrieves a ProductUsage entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieving a product usage with its id

Request

```
GET /productUsage/83c780b5-517c-4e28-9924-625e3fc7b341
Content-Type: application/json
```

Response

```
200
{
```

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```
  "@type": "ProductUsage",
  "href": "https://mycsp.com/tmf-
api/productUsageManagement/v5/productUsage/83c780b5-517c-4e28-9924-625e3fc7b341",
  "id": "83c780b5-517c-4e28-9924-625e3fc7b341",
  "creationDate": "2023-09-11T14:35:05Z",
  "lastUpdate": "2023-09-11T14:38:00Z",
  "status": "rated",
  "usageType": "InternationalVoice",
  "usageCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "name": "originatingCountryCode",
      "value": "FR"
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationCountryCode",
      "value": "GB"
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationPhoneNumber",
      "value": "+447092734896"
    },
    {
      "@type": "StringCharacteristic",
      "name": "originatingPhoneNumber",
      "value": "+33939487561"
    },
    {
      "@type": "StringCharacteristic",
      "name": "callType",
      "value": "outgoing"
    },
    {
      "@type": "IntegerCharacteristic",
      "name": "duration",
      "value": 300
    },
    {
      "@type": "StringCharacteristic",
      "name": "startDateTime",
      "valueType": "date-time",
      "value": "2023-09-11T14:30:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "endDateTime",
      "valueType": "date-time",
      "value": "2023-09-11T14:35:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "callStatus",
      "value": "completed"
    }
  ],
  "usageProductPrice": [
    {
      "@type": "ProductPrice",
      "price": {
        "@type": "Price",
        "dutyFreeAmount": {
```


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```
          "unit": "EUR",
          "value": 2.5
        },
        "taxRate": 20
      },
      "productOfferingPrice": {
        "@type": "ProductOfferingPriceRef",
        "@referredType": "ProductOfferingPrice",
        "href": "https://mycsp.com/tmf-
api/productCatalogManagement/v5/productOffering/419ce511-dbb9-4753-8155-
3391b9998cd7",
        "id": "419ce511-dbb9-4753-8155-3391b9998cd7"
      },
      "priceType": "usage"
    }
  ],
  "serviceUsage": [
    {
      "@type": "ServiceUsageRef",
      "@referredType": "ServiceUsage",
      "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsage/472999bf-1aeb-48e6-b954-1303c2c7939b",
      "id": "472999bf-1aeb-48e6-b954-1303c2c7939b"
    }
  ],
  "product": {
    "@type": "ProductRef",
    "@referredType": "Product",
    "href": "https://mycsp.com/tmf-api/productInventory/v5/product/05c89397-6984-
4c8e-9030-7cf092754f18",
    "id": "05c89397-6984-4c8e-9030-7cf092754f18",
    "name": "FixCommunication"
  },
  "usageSpecification": {
    "@type": "ProductUsageSpecificationRef",
    "@referredType": "ProductUsageSpecification",
    "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
    "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
    "name": "InternationalVoice",
    "version": "2.5"
  }
}
```


List or find ProductUsage objects

`GET /productUsage?fields=…`

Description

This operation list ProductUsage entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieving a list of product usage of usageType 'InternationalVoice' with a selection of attributes.

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Request

```
GET
/productUsage?fields=status,usageType,creationDate,usageSpecification,product,usageC
haracteristic&usageType=InternationalVoice
Content-Type: application/json
```

Response

```
200
[
{
  "@type": "ProductUsage",
  "href": "https://mycsp.com/tmf-
api/productUsageManagement/v5/productUsage/83c780b5-517c-4e28-9924-625e3fc7b341",
  "id": "83c780b5-517c-4e28-9924-625e3fc7b341",
  "creationDate": "2023-09-11T14:35:05Z",
  "status": "rated",
  "usageType": "InternationalVoice",
  "usageCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "name": "originatingCountryCode",
      "value": "FR"
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationCountryCode",
      "value": "GB"
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationPhoneNumber",
      "value": "+447092734896"
    },
    {
      "@type": "StringCharacteristic",
      "name": "originatingPhoneNumber",
      "value": "+33939487561"
    },
    {
      "@type": "StringCharacteristic",
      "name": "callType",
      "value": "outgoing"
    },
    {
      "@type": "IntegerCharacteristic",
      "name": "duration",
      "value": 300
    },
    {
      "@type": "StringCharacteristic",
      "name": "startDateTime",
      "valueType": "date-time",
      "value": "2023-09-11T14:30:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "endDateTime",
```

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```
      "valueType": "date-time",
      "value": "2023-09-11T14:35:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "callStatus",
      "value": "completed"
    }
  ],
  "product": {
    "@type": "ProductRef",
    "@referredType": "Product",
    "href": "https://mycsp.com/tmf-api/productInventory/v5/product/05c89397-6984-
4c8e-9030-7cf092754f18",
    "id": "05c89397-6984-4c8e-9030-7cf092754f18",
    "name": "FixCommunication"
  },
  "usageSpecification": {
    "@type": "ProductUsageSpecificationRef",
    "@referredType": "ProductUsageSpecification",
    "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
    "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
    "name": "InternationalVoice",
    "version": "2.5"
  }
},
  {
    "@type": "ProductUsage",
    "href": "https://mycsp.com/tmf-
api/productUsageManagement/v5/productUsage/bc610b6a-1fb3-4b31-8750-60788426554e",
    "id": "bc610b6a-1fb3-4b31-8750-60788426554e",
    "creationDate": "2023-09-11T12:00:22Z",
    "status": "rated",
    "usageType": "InternationalVoice",
    "usageCharacteristic": [
      {
        "@type": "StringCharacteristic",
        "name": "originatingCountryCode",
        "value": "FR"
      },
      {
        "@type": "StringCharacteristic",
        "name": "destinationCountryCode",
        "value": "GB"
      },
      {
        "@type": "StringCharacteristic",
        "name": "destinationPhoneNumber",
        "value": "+447123456789"
      },
      {
        "@type": "StringCharacteristic",
        "name": "originatingPhoneNumber",
        "value": "+33987654321"
      },
      {
        "@type": "StringCharacteristic",
        "name": "callType",
        "value": "outgoing"
      },
      {
```


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```
        "@type": "IntegerCharacteristic",
        "name": "duration",
        "value": 3600
      },
      {
        "@type": "StringCharacteristic",
        "name": "startDateTime",
        "valueType": "date-time",
        "value": "2023-09-11T11:00:13Z"
      },
      {
        "@type": "StringCharacteristic",
        "name": "endDateTime",
        "valueType": "date-time",
        "value": "2023-09-11T12:00:13Z"
      },
      {
        "@type": "StringCharacteristic",
        "name": "callStatus",
        "value": "completed"
      }
    ],
  "product": {
    "@type": "ProductRef",
    "@referredType": "Product",
    "href": "https://mycsp.com/tmf-api/productInventory/v5/product/05c89397-6984-
4c8e-9030-7cf092754f18",
    "id": "05c89397-6984-4c8e-9030-7cf092754f18",
    "name": "FixCommunication"
  },
  "usageSpecification": {
    "@type": "ProductUsageSpecificationRef",
    "@referredType": "ProductUsageSpecification",
    "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
    "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
    "name": "InternationalVoice",
    "version": "2.5"
  }
  }
]
```


Creates a ProductUsage

POST /productUsage?fields=…

Description

This operation creates a ProductUsage entity.

Mandatory Attributes

Mandatory Attributes	Rule
product	
usageSpecification	
@type	M

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Usage samples

Creating a product usage resource.

Request

```
POST /productUsage
Content-Type: application/json
{
  "@type": "ProductUsage",
  "usageType": "InternationalVoice",
  "usageCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "name": "originatingCountryCode",
      "value": "FR"
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationCountryCode",
      "value": "GB"
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationPhoneNumber",
      "value": "+447092734896"
    },
    {
      "@type": "StringCharacteristic",
      "name": "originatingPhoneNumber",
      "value": "+33939487561"
    },
    {
      "@type": "StringCharacteristic",
      "name": "callType",
      "value": "outgoing"
    },
    {
      "@type": "IntegerCharacteristic",
      "name": "duration",
      "value": 300
    },
    {
      "@type": "StringCharacteristic",
      "name": "startDateTime",
      "valueType": "date-time",
      "value": "2023-09-11T14:30:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "endDateTime",
      "valueType": "date-time",
      "value": "2023-09-11T14:35:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "callStatus",
      "value": "completed"
    }
  ],
  "usageProductPrice": [
```

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```
    {
      "@type": "ProductPrice",
      "price": {
        "@type": "Price",
        "dutyFreeAmount": {
          "unit": "EUR",
          "value": 2.5
        },
        "taxRate": 20
      },
      "productOfferingPrice": {
        "@type": "ProductOfferingPriceRef",
        "@referredType": "ProductOfferingPrice",
        "href": "https://mycsp.com/tmf-
api/productCatalogManagement/v5/productOffering/419ce511-dbb9-4753-8155-
3391b9998cd7",
        "id": "419ce511-dbb9-4753-8155-3391b9998cd7"
      },
      "priceType": "usage"
    }
  ],
  "serviceUsage": [
    {
      "@type": "ServiceUsageRef",
      "@referredType": "ServiceUsage",
      "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsage/472999bf-1aeb-48e6-b954-1303c2c7939b",
      "id": "472999bf-1aeb-48e6-b954-1303c2c7939b"
    }
  ],
  "product": {
    "@type": "ProductRef",
    "@referredType": "Product",
    "href": "https://api.csp.com/tmf-api/product/05c89397-6984-4c8e-9030-
7cf092754f18",
    "id": "05c89397-6984-4c8e-9030-7cf092754f18",
    "name": "FixCommunication"
  },
  "usageSpecification": {
    "@type": "ProductUsageSpecificationRef",
    "@referredType": "ProductUsageSpecification",
    "href": "https://api.csp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
    "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
    "name": "InternationalVoice",
    "version": "2.5"
  }
}
```


Response


```
201
{
  "@type": "ProductUsage",
  "href": "https://mycsp.com/tmf-
api/productUsageManagement/v5/productUsage/83c780b5-517c-4e28-9924-625e3fc7b341",
  "id": "83c780b5-517c-4e28-9924-625e3fc7b341",
  "creationDate": "2023-09-11T14:35:05Z",
  "status": "initialized",
```


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```
  "usageType": "InternationalVoice",
  "usageCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "name": "originatingCountryCode",
      "value": "FR"
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationCountryCode",
      "value": "GB"
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationPhoneNumber",
      "value": "+447092734896"
    },
    {
      "@type": "StringCharacteristic",
      "name": "originatingPhoneNumber",
      "value": "+33939487561"
    },
    {
      "@type": "StringCharacteristic",
      "name": "callType",
      "value": "outgoing"
    },
    {
      "@type": "IntegerCharacteristic",
      "name": "duration",
      "value": 300
    },
    {
      "@type": "StringCharacteristic",
      "name": "startDateTime",
      "valueType": "date-time",
      "value": "2023-09-11T14:30:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "endDateTime",
      "valueType": "date-time",
      "value": "2023-09-11T14:35:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "callStatus",
      "value": "completed"
    }
  ],
  "serviceUsage": [
    {
      "@type": "ServiceUsageRef",
      "@referredType": "ServiceUsage",
      "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsage/472999bf-1aeb-48e6-b954-1303c2c7939b",
      "id": "472999bf-1aeb-48e6-b954-1303c2c7939b"
    }
  ],
  "product": {
    "@type": "ProductRef",
    "@referredType": "Product",
    "href": "https://mycsp.com/tmf-api/productInventory/v5/product/05c89397-6984-
```


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```
4c8e-9030-7cf092754f18",
    "id": "05c89397-6984-4c8e-9030-7cf092754f18",
    "name": "FixCommunication"
  },
  "usageSpecification": {
    "@type": "ProductUsageSpecificationRef",
    "@referredType": "ProductUsageSpecification",
    "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
    "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
    "name": "InternationalVoice",
    "version": "2.5"
  }
}
```


Updates partially a ProductUsage

`PATCH /productUsage/{id}?fields=…`

Description

This operation allows partial updates of a ProductUsage entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
creationDate	
href	
id	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
externalIdentifier	
lastUpdate	
product	
relatedParty	
serviceUsage	
status	
usageCharacteristic	

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Patchable Attributes	Rule
usageProductPrice	
usageSpecification	
usageType	

Usage samples

Product Usage update example with json content type

Request

```
PATCH /productUsage/83c780b5-517c-4e28-9924-625e3fc7b341
Content-Type: application/json
{
  "@type": "ProductUsage",
  "status": "superseded"
}
```

Response

```
200
{
  "@type": "ProductUsage",
  "href": "https://mycsp.com/tmf-
api/productUsageManagement/v5/productUsage/83c780b5-517c-4e28-9924-625e3fc7b341",
  "id": "83c780b5-517c-4e28-9924-625e3fc7b341",
  "creationDate": "2023-09-11T14:35:05Z",
  "lastUpdate": "2023-09-11T14:40:00Z",
  "status": "superseded",
  "usageType": "InternationalVoice",
  "usageCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "name": "originatingCountryCode",
      "value": "FR"
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationCountryCode",
      "value": "GB"
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationPhoneNumber",
      "value": "+447092734896"
    },
    {
      "@type": "StringCharacteristic",
      "name": "originatingPhoneNumber",
      "value": "+33939487561"
    },
    {
      "@type": "StringCharacteristic",
      "name": "callType",
      "value": "outgoing"
```

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```
    },
    {
      "@type": "IntegerCharacteristic",
      "name": "duration",
      "value": 300
    },
    {
      "@type": "StringCharacteristic",
      "name": "startDateTime",
      "valueType": "date-time",
      "value": "2023-09-11T14:30:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "endDateTime",
      "valueType": "date-time",
      "value": "2023-09-11T14:35:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "callStatus",
      "value": "completed"
    }
  ],
  "usageProductPrice": [
    {
      "@type": "ProductPrice",
      "price": {
        "@type": "Price",
        "dutyFreeAmount": {
          "unit": "EUR",
          "value": 2.6
        },
        "taxRate": 20
      },
      "productOfferingPrice": {
        "@type": "ProductOfferingPriceRef",
        "@referredType": "ProductOfferingPrice",
        "href": "https://mycsp.com/tmf-
api/productCatalogManagement/v5/productOffering/419ce511-dbb9-4753-8155-
3391b9998cd7",
        "id": "419ce511-dbb9-4753-8155-3391b9998cd7"
      },
      "priceType": "usage"
    }
  ],
  "serviceUsage": [
    {
      "@type": "ServiceUsageRef",
      "@referredType": "ServiceUsage",
      "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsage/472999bf-1aeb-48e6-b954-1303c2c7939b",
      "id": "472999bf-1aeb-48e6-b954-1303c2c7939b"
    }
  ],
  "product": {
    "@type": "ProductRef",
    "@referredType": "Product",
    "href": "https://mycsp.com/tmf-api/productInventory/v5/product/05c89397-6984-
4c8e-9030-7cf092754f18",
    "id": "05c89397-6984-4c8e-9030-7cf092754f18",
    "name": "FixCommunication"
  },
```


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```
  "usageSpecification": {
    "@type": "ProductUsageSpecificationRef",
    "@referredType": "ProductUsageSpecification",
    "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
    "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
    "name": "InternationalVoice",
    "version": "2.5"
  }
}
```

Product Usage update example with merge-patch+json content type

Request

```
PATCH /productUsage/83c780b5-517c-4e28-9924-625e3fc7b341
Content-Type: application/merge-patch+json
{
  "@type": "ProductUsage",
  "status": "superseded"
}
```

Response

```
200
{
  "@type": "ProductUsage",
  "href": "https://mycsp.com/tmf-
api/productUsageManagement/v5/productUsage/83c780b5-517c-4e28-9924-625e3fc7b341",
  "id": "83c780b5-517c-4e28-9924-625e3fc7b341",
  "creationDate": "2023-09-11T14:35:05Z",
  "lastUpdate": "2023-09-11T14:40:00Z",
  "status": "superseded",
  "usageType": "InternationalVoice",
  "usageCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "name": "originatingCountryCode",
      "value": "FR"
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationCountryCode",
      "value": "GB"
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationPhoneNumber",
      "value": "+447092734896"
    },
    {
      "@type": "StringCharacteristic",
      "name": "originatingPhoneNumber",
      "value": "+33939487561"
    },
    {
```

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```
      "@type": "StringCharacteristic",
      "name": "callType",
      "value": "outgoing"
    },
    {
      "@type": "IntegerCharacteristic",
      "name": "duration",
      "value": 300
    },
    {
      "@type": "StringCharacteristic",
      "name": "startDateTime",
      "valueType": "date-time",
      "value": "2023-09-11T14:30:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "endDateTime",
      "valueType": "date-time",
      "value": "2023-09-11T14:35:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "callStatus",
      "value": "completed"
    }
  ],
  "usageProductPrice": [
    {
      "@type": "ProductPrice",
      "price": {
        "@type": "Price",
        "dutyFreeAmount": {
          "unit": "EUR",
          "value": 2.6
        },
        "taxRate": 20
      },
      "productOfferingPrice": {
        "@type": "ProductOfferingPriceRef",
        "@referredType": "ProductOfferingPrice",
        "href": "https://mycsp.com/tmf-
api/productCatalogManagement/v5/productOffering/419ce511-dbb9-4753-8155-
3391b9998cd7",
        "id": "419ce511-dbb9-4753-8155-3391b9998cd7"
      },
      "priceType": "usage"
    }
  ],
  "serviceUsage": [
    {
      "@type": "ServiceUsageRef",
      "@referredType": "ServiceUsage",
      "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsage/472999bf-1aeb-48e6-b954-1303c2c7939b",
      "id": "472999bf-1aeb-48e6-b954-1303c2c7939b"
    }
  ],
  "product": {
    "@type": "ProductRef",
    "@referredType": "Product",
    "href": "https://mycsp.com/tmf-api/productInventory/v5/product/05c89397-6984-
4c8e-9030-7cf092754f18",
```


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```
    "id": "05c89397-6984-4c8e-9030-7cf092754f18",
    "name": "FixCommunication"
  },
  "usageSpecification": {
    "@type": "ProductUsageSpecificationRef",
    "@referredType": "ProductUsageSpecification",
    "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
    "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
    "name": "InternationalVoice",
    "version": "2.5"
  }
}
```

Product Usage update example with json-patch+json content type

Request

```
PATCH /productUsage/83c780b5-517c-4e28-9924-625e3fc7b341
Content-Type: application/json-patch+json
[
  {
    "op": "replace",
    "path": "/status",
    "value": "superseded"
  }
]
```

Response

```
200
{
  "@type": "ProductUsage",
  "href": "https://mycsp.com/tmf-
api/productUsageManagement/v5/productUsage/83c780b5-517c-4e28-9924-625e3fc7b341",
  "id": "83c780b5-517c-4e28-9924-625e3fc7b341",
  "creationDate": "2023-09-11T14:35:05Z",
  "lastUpdate": "2023-09-11T14:40:00Z",
  "status": "superseded",
  "usageType": "InternationalVoice",
  "usageCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "name": "originatingCountryCode",
      "value": "FR"
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationCountryCode",
      "value": "GB"
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationPhoneNumber",
      "value": "+447092734896"
    },
```

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```
    {
      "@type": "StringCharacteristic",
      "name": "originatingPhoneNumber",
      "value": "+33939487561"
    },
    {
      "@type": "StringCharacteristic",
      "name": "callType",
      "value": "outgoing"
    },
    {
      "@type": "IntegerCharacteristic",
      "name": "duration",
      "value": 300
    },
    {
      "@type": "StringCharacteristic",
      "name": "startDateTime",
      "valueType": "date-time",
      "value": "2023-09-11T14:30:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "endDateTime",
      "valueType": "date-time",
      "value": "2023-09-11T14:35:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "callStatus",
      "value": "completed"
    }
  ],
  "usageProductPrice": [
    {
      "@type": "ProductPrice",
      "price": {
        "@type": "Price",
        "dutyFreeAmount": {
          "unit": "EUR",
          "value": 2.6
        },
        "taxRate": 20
      },
      "productOfferingPrice": {
        "@type": "ProductOfferingPriceRef",
        "@referredType": "ProductOfferingPrice",
        "href": "https://mycsp.com/tmf-
api/productCatalogManagement/v5/productOffering/419ce511-dbb9-4753-8155-
3391b9998cd7",
        "id": "419ce511-dbb9-4753-8155-3391b9998cd7"
      },
      "priceType": "usage"
    }
  ],
  "serviceUsage": [
    {
      "@type": "ServiceUsageRef",
      "@referredType": "ServiceUsage",
      "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsage/472999bf-1aeb-48e6-b954-1303c2c7939b",
      "id": "472999bf-1aeb-48e6-b954-1303c2c7939b"
    }
```


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```
  ],
  "product": {
    "@type": "ProductRef",
    "@referredType": "Product",
    "href": "https://mycsp.com/tmf-api/productInventory/v5/product/05c89397-6984-
4c8e-9030-7cf092754f18",
    "id": "05c89397-6984-4c8e-9030-7cf092754f18",
    "name": "FixCommunication"
  },
  "usageSpecification": {
    "@type": "ProductUsageSpecificationRef",
    "@referredType": "ProductUsageSpecification",
    "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
    "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
    "name": "InternationalVoice",
    "version": "2.5"
  }
}
```


Product Usage update example with json-patch-query+json content type

Request

```
PATCH /productUsage/83c780b5-517c-4e28-9924-625e3fc7b341
Content-Type: application/json-patch-query+json
[
  {
    "op": "replace",
    "path": "/usageCharacteristic/value?name=originatingPhoneNumber",
    "value": "+33939487562"
  }
]
```

Response


```
200
{
  "@type": "ProductUsage",
  "href": "https://mycsp.com/tmf-
api/productUsageManagement/v5/productUsage/83c780b5-517c-4e28-9924-625e3fc7b341",
  "id": "83c780b5-517c-4e28-9924-625e3fc7b341",
  "creationDate": "2023-09-11T14:35:05Z",
  "lastUpdate": "2023-09-11T14:55:00Z",
  "status": "rated",
  "usageType": "InternationalVoice",
  "usageCharacteristic": [
    {
      "@type": "StringCharacteristic",
      "name": "originatingCountryCode",
      "value": "FR"
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationCountryCode",
      "value": "GB"
```


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```
    },
    {
      "@type": "StringCharacteristic",
      "name": "destinationPhoneNumber",
      "value": "+447092734896"
    },
    {
      "@type": "StringCharacteristic",
      "name": "originatingPhoneNumber",
      "value": "+33939487562"
    },
    {
      "@type": "StringCharacteristic",
      "name": "callType",
      "value": "outgoing"
    },
    {
      "@type": "IntegerCharacteristic",
      "name": "duration",
      "value": 300
    },
    {
      "@type": "StringCharacteristic",
      "name": "startDateTime",
      "valueType": "date-time",
      "value": "2023-09-11T14:30:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "endDateTime",
      "valueType": "date-time",
      "value": "2023-09-11T14:35:00Z"
    },
    {
      "@type": "StringCharacteristic",
      "name": "callStatus",
      "value": "completed"
    }
  ],
  "usageProductPrice": [
    {
      "@type": "ProductPrice",
      "price": {
        "@type": "Price",
        "dutyFreeAmount": {
          "unit": "EUR",
          "value": 2.5
        },
        "taxRate": 20
      },
      "productOfferingPrice": {
        "@type": "ProductOfferingPriceRef",
        "@referredType": "ProductOfferingPrice",
        "href": "https://mycsp.com/tmf-
api/productCatalogManagement/v5/productOffering/419ce511-dbb9-4753-8155-
3391b9998cd7",
        "id": "419ce511-dbb9-4753-8155-3391b9998cd7"
      },
      "priceType": "usage"
    }
  ],
  "serviceUsage": [
    {
```


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```
      "@type": "ServiceUsageRef",
      "@referredType": "ServiceUsage",
      "href": "https://mycsp.com/tmf-
api/serviceUsageManagement/v5/serviceUsage/472999bf-1aeb-48e6-b954-1303c2c7939b",
      "id": "472999bf-1aeb-48e6-b954-1303c2c7939b"
    }
  ],
  "product": {
    "@type": "ProductRef",
    "@referredType": "Product",
    "href": "https://mycsp.com/tmf-api/productInventory/v5/product/05c89397-6984-
4c8e-9030-7cf092754f18",
    "id": "05c89397-6984-4c8e-9030-7cf092754f18",
    "name": "FixCommunication"
  },
  "usageSpecification": {
    "@type": "ProductUsageSpecificationRef",
    "@referredType": "ProductUsageSpecification",
    "href": "https://mycsp.com/tmf-
api/productUsageCatalogManagement/v5/productUsageSpecification/c2d6642b-59be-4cd1-
800d-f3920dd4bc7f",
    "id": "c2d6642b-59be-4cd1-800d-f3920dd4bc7f",
    "name": "InternationalVoice",
    "version": "2.5"
  }
}
```


Deletes a ProductUsage

DELETE /productUsage/{id}

Description

This operation deletes a ProductUsage entity.

Usage samples

Requesting a product usage to be deleted.

Request

```
DELETE /productUsage/83c780b5-517c-4e28-9924-625e3fc7b341
Content-Type: application/json
```

Response

```
204
```

Operations on ExportJob

List or find ExportJob objects

GET /exportJob?fields=…

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Description

This operation list ExportJob entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieving a list of export job of status 'Succeeded'.

Request

```
GET /exportJob?status=Succeeded
Content-Type: application/json
```

Response

```
200
[
  {
  "@type": "ExportJob",
  "href": "https://mycsp.com/tmf-api/productUsageManagement/v5/exportJob/aa1d7102-
a2de-4e50-abc5-15cddac9c5d4",
  "id": "aa1d7102-a2de-4e50-abc5-15cddac9c5d4",
  "creationDate": "2023-10-01T00:00:20Z",
  "completionDate": "2023-10-01T00:02:30Z",
  "status": "Succeeded",
  "path": "/productUsage",
  "query": "status=validated,rated,notRated,superseded&creationDate.gte=2023-09-
10T00:00:00Z&creationDate.lt=2023-10-10T00:00:00Z",
    "contentType": "application/gzip",
    "url": "sftp://sftp.mycsp.com/productUsageManagement/fzohfozg567gs5fg7dg3.gzip"
},
{
  "@type": "ExportJob",
  "href": "https://mycsp.com/tmf-api/productUsageManagement/v5/exportJob/e72d56d5-
f5d7-473b-afe5-1e89eeed012b",
  "id": "e72d56d5-f5d7-473b-afe5-1e89eeed012b",
  "creationDate": "2023-11-01T00:00:20Z",
  "completionDate": "2023-11-01T00:02:30Z",
  "status": "Succeeded",
  "path": "/productUsage",
  "query": "status=validated,rated,notRated,superseded&creationDate.gte=2023-10-
10T00:00:00Z&creationDate.lt=2023-11-10T00:00:00Z",
    "contentType": "application/gzip",
    "url": "sftp://sftp.mycsp.com/productUsageManagement/jiozjoe897cdjzd98.gzip"
}
]
```

Retrieves a ExportJob by ID

`GET /exportJob/{id}?fields=…`

Description

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This operation retrieves an ExportJob entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieving an export job with its id

Request

```
GET /exportJob/aa1d7102-a2de-4e50-abc5-15cddac9c5d4
Content-Type: application/json
```

Response

```
200
{
  "@type": "ExportJob",
  "href": "https://mycsp.com/tmf-api/productUsageManagement/v5/exportJob/aa1d7102-
a2de-4e50-abc5-15cddac9c5d4",
  "id": "aa1d7102-a2de-4e50-abc5-15cddac9c5d4",
  "creationDate": "2023-10-01T00:00:20Z",
  "completionDate": "2023-10-01T00:02:30Z",
  "status": "Succeeded",
  "path": "/productUsage",
  "query": "status=validated,rated,notRated,superseded&creationDate.gte=2023-09-
10T00:00:00Z&creationDate.lt=2023-10-10T00:00:00Z",
    "contentType": "application/gzip",
    "url": "sftp://sftp.mycsp.com/productUsageManagement/fzohfozg567gs5fg7dg3.gzip"
}
```

Initiate a task for ExportJob

`POST /exportJob?fields=…`

Description

This operation creates an ExportJob entity.

Mandatory Attributes

Mandatory Attributes	Rule
@type	M

Usage samples

Requesting an export job task to be launched.

Request

```
POST /exportJob
```

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```
Content-Type: application/json
{
  "@type": "ExportJob",
  "path": "/productUsage",
  "query": "status=validated,rated,notRated,superseded&creationDate.gte=2023-09-
10T00:00:00Z&creationDate.lt=2023-10-10T00:00:00Z",
    "contentType": "application/gzip",
    "url": "sftp://sftp.mycsp.com/productUsageManagement/fzohfozg567gs5fg7dg3.gzip"
}
```

Response

```
201
{
  "@type": "ExportJob",
  "href": "https://mycsp.com/tmf-api/productUsageManagement/v5/exportJob/aa1d7102-
a2de-4e50-abc5-15cddac9c5d4",
  "id": "aa1d7102-a2de-4e50-abc5-15cddac9c5d4",
  "creationDate": "2023-10-01T00:00:20Z",
  "status": "Running",
  "path": "/productUsage",
  "query": "status=validated,rated,notRated,superseded&creationDate.gte=2023-09-
10T00:00:00Z&creationDate.lt=2023-10-10T00:00:00Z",
    "contentType": "application/gzip",
    "url": "sftp://sftp.mycsp.com/productUsageManagement/fzohfozg567gs5fg7dg3.gzip"
}
```

Deletes a ExportJob

`DELETE /exportJob/{id}`

Description

This operation deletes an ExportJob entity.

Usage samples

Requesting an export job task to be deleted.

Request

```
DELETE /exportJob/aa1d7102-a2de-4e50-abc5-15cddac9c5d4
Content-Type: application/json
```

Response

```
204
```

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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

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Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

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```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release Notes

V4.0.0 vs v5.0.0

general changes

- common changes due to v5 tooling assets (OAS, User Guide, Conformance Profile…) generation
- usageSpecification endpoint removed, moved to TMF767 Product Usage Catalog Management API, same for usageSpecification related events
- endpoint /usage renamed into /productUsage. Note that serviceUsage and resourceUsage endpoints are managed in specific API TMF727 and TMF771
- usageXXX events removed (no more events)
- productUsage lifecycle changed to fit better the usage definition in Product domain
- addition of /exportJob resource to allow export of product usage data in a file, and exportJob events to receive corresponding notifications
- User Guide updated with excerpts from Information Framework and TMFS005 Use Case
- Conformance Profile updated accordingly to these changes

schemas changes

- productUsage : usageDate renamed into creationDate to follow general guidelines
- productUsage : lastUpdate added to have the latest update date of the product usage
- productUsage : externalIdentifier added to allow an array of external identifiers related to the product usage
- productUsage : status changed from enum to open string with enum proposal to allow more flexibility
- productUsage : usageCharacteristic uses generic Characteristic schema instead of UsageCharacteristic schema, to follow general guidelines
- productUsage : serviceUsage added to give the identifier of the service usage(s) that participated in the creation of the product usage
- productUsage : relatedParty uses RelatedPartyRefOrPartyRoleRef schema instead of RelatedParty schema to follow general guidelines
- productUsage : ratedProductUsage renamed into usageProductPrice to express this is a price
- productUsage : usageProductPrice uses common schema ProductPrice instead of RatedProductUsage schema
- productUsage : product reference is in the productUsage root instead of being in the usage price attribute for better clarity
- productUsage : usageSpecification uses ProductUsageSpecificationRef schema instead of UsageSpecificationRef schema, for better precision on the targetted schema

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- productUsage : usageSpecification, usageCharacteristic, product and usageProductPrice.productOfferingPrice are required for the ProductUsage_FVO schema

Release History

Release Number	Date	Release led by:	Description
V4.0.0	July 2021	Kamal Maghsoudlou / Ericsson, Kevin McDonnell / Huawei	1st release of the Usage API
v4.1.0 preview		March 2024	
v5.0.0 preview	June 2024	Sergei Lukin / Deutsche Telekom	migrated to OAS 3.0
v5.0.0	November 2025	Olivier Arnaud / Orange	final upgrades for v5 migration

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